

James E. Kajdasz, Ph.D.

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SUMMARY

Senior Data Scientist specializing in fraud detection, identity risk, and adversarial behavior modeling, with 20+ years building analytics and decision-support systems in high-stakes environments. Developed fraud detection systems that identified more than \$1B in fraudulent claims exposure across public-benefits programs and generated 100+ high-value investigative leads involving synthetic identity fraud, mule networks, and coordinated fraud rings. Expertise in identity resolution, graph analytics, feature engineering, and risk scoring across large-scale relational datasets. Combines deep technical expertise in Python, machine learning, and applied statistics with extensive experience in intelligence analysis, uncertainty reasoning, and decision-making under adversarial conditions.

TECHNICAL SKILLS

Languages: Python (Advanced), SQL (Working Proficiency)

Machine Learning & Analytics: Fraud Detection, Identity Resolution, Entity Resolution, Feature Engineering, Risk Scoring, Graph Analytics, Link Analysis, Statistical Modeling, Logistic Regression, Classification, CNNs, Forecasting, Bayesian Analysis

Cloud & Infrastructure: AWS (EC2, S3, boto3), Snowflake

Tools & Developer Workflow: Streamlit, Cytoscape, Git, VS Code, UV, Claude Code, GitHub Copilot, Conda, pip

Modern AI: Agentic AI, RAG, LangChain, LangGraph, CrewAI, Vector Databases

PROFESSIONAL EXPERIENCE

Senior Data Scientist — Thomson Reuters Special Services (TRSS) | 2022–Present

- Led fraud analytics strategy for a California public-benefits fraud engagement focused on unemployment insurance (UI), disability insurance (DI), and hospice fraud, identifying synthetic identity theft, mule networks, coordinated fraud rings, and organized criminal fraud.
- Selected as the only data scientist retained across the full lifecycle of the engagement, which expanded from a 3-month proof-of-concept with 6 data scientists and 6 analysts into a 3-year production contract.
- Developed over 100 high-value fraud leads contributing to identification of more than \$1B in fraudulent claims exposure; contributed directly to every major lead developed during the contract.
- Architected and built PyEDD, an installable Python fraud analytics library (pip/conda) using object-oriented design and Git versioning that automated claims extraction, third-party enrichment, feature engineering, graph-analysis exports, and investigative reporting.
- Built Python workflows that generated SQL queries against Snowflake claims tables spanning approximately 24 relational datasets, enabling rapid investigation of claims linked by phone number, email, IP address, SSN, and street address.
- Integrated claims data with external fraud intelligence sources including IPQS, Smarty, public records, and proprietary datasets to perform identity resolution and enrich claims with geolocation, linkage, and fraud-risk signals.
- Engineered high-value fraud features including claimant concentration by address/email/IP/phone, vendor fraud scores, criminal history indicators, and relational linkage metrics used to prioritize investigative leads.
- Designed and implemented “Missing Man Analysis,” a scalable address-linkage workflow capable of processing thousands of addresses to identify latent fraud operators who never directly submitted claims but appeared across multiple suspicious claim clusters.
- Developed graph-analysis pipelines that generated node/edge ingestion files for Cytoscape, enabling visual discovery of coordinated fraud rings, shared infrastructure, and hidden entity relationships.
- Used graph analytics and relational linkage analysis to uncover complex fraud rings, including one case linking four high-risk addresses through shared infrastructure to an unmasked IP in Juarez, Mexico, ultimately identifying a central suspect associated with organized criminal activity.
- Built a Streamlit self-service analytics platform that enabled non-technical analysts to search claims by address, phone, email, SSN, and IP without SQL, dramatically reducing analyst dependency on data scientists for routine investigation.

Machine Learning Engineer — Electromagnetic Systems, Inc. (EMSI) | 2020–2022

- Developed machine learning solutions for automatic target recognition (ATR) on large-scale government sensor imagery, supporting computer vision applications in operational environments.
- Built and executed ML workflows in AWS GovCloud using EC2, S3, and boto3 to manage data ingestion, storage, experimentation, and model execution.
- Performed hyperparameter tuning and model performance optimization for convolutional neural network (CNN) architectures to improve target-detection accuracy under noisy and challenging sensor conditions.
- Designed ETL pipelines to ingest, clean, normalize, and standardize heterogeneous imagery datasets sourced from multiple government agencies.
- Processed and validated training datasets with inconsistent labeling formats, including point-based and bounding-box annotations with varying quality and precision.
- Improved training data quality by reviewing imagery, standardizing annotations, and resolving inconsistencies across data providers to create reliable inputs for model training and evaluation.

Chief, Operations Assessment — U.S. Army Special Operations | Doha, Qatar | 2017

- Led operations assessment for U.S. information campaigns across approximately nine countries in the Middle East and Central Asia, including Gulf states, Lebanon, and Afghanistan, focused on reducing support for ISIS and other extremist networks.
- Developed psychometrically informed measurement frameworks that transformed ambiguous strategic objectives into objective performance indicators for leadership decision-making.
- Designed longitudinal scoring methodologies using cognitive and behavioral indicators to evaluate progress on information objectives such as reducing insurgent support and sectarian tensions.
- Leveraged large-scale national survey data from conflict and non-conflict regions to measure population attitudes, expectations, and behaviors over time.
- Delivered assessment reports that became a preferred framework for campaign evaluation, influencing messaging priorities and strategic planning across Army psychological operations teams.

Research Deputy Director & Assistant Professor — U.S. Air Force Academy | 2014–2018

- Served as Deputy Director of an Air Force-level R&D institute, leading daily operations and directly managing a multidisciplinary team of civilian, contractor, and military personnel.
- Served as Air Force Research Laboratory (AFRL) liaison to USAFA's \$38M+ research enterprise, aligning institutional research efforts with Air Force strategic priorities and operational needs.
- Evaluated approximately \$30M in federal research proposals and provided strategic recommendations influencing prioritization of advanced research funding, including support to IARPA-related initiatives.
- Directed multidisciplinary technical programs spanning cyber operations, information technology, unmanned systems, and command-and-control research.
- Evaluated feasibility of counter-UAS research initiatives and identified critical infrastructure constraints that limited viable experimentation at USAFA.
- Successfully redirected institutional investment toward higher-value unmanned systems research by securing funding and leading end-to-end development of a \$500K UAS command-and-control center focused on swarm operations and remotely piloted aircraft systems.
- Managed multiple technical infrastructure projects including a \$300K data wall installation, \$124K network modernization effort, and \$65K infrastructure upgrade.
- Taught psychology, leadership theory, and research methods to future Air Force officers.

Faculty, Analytic Methodology & IRB Chair — National Intelligence University | 2010–2014

- Taught graduate and undergraduate courses in intelligence, analytic methodology, globalization, economics, joint planning, and strategy to 700+ military and civilian intelligence professionals across DIA, NSA, and NGA campuses.
- Developed and taught advanced coursework in structured analytic techniques, cognitive bias mitigation, Analysis of Competing Hypotheses (ACH), argument mapping, intelligence reasoning, forecasting, Bayesian reasoning, signal detection theory, utility theory, linear modeling, research design, and decision-making under uncertainty.
- Mentored intelligence professionals and chaired multiple master's theses involving high-impact research in intelligence analysis, critical thinking, and national security decision-making.
- Recognized intelligence analytics expert; invited to advise ODNI and other intelligence-community organizations on critical thinking, analytic rigor, and advanced research methods.
- Chaired NIU's Institutional Review Board, reviewing 400+ research proposals and ensuring compliance with federal, DoD, and intelligence-community regulations, including classified research.
- Served as NIU representative to IARPA funding committees, helping prioritize multimillion-dollar research investments supporting intelligence-community innovation.
- Published and presented research on intelligence analysis, prediction markets, crowd-based analysis, and decision-making under uncertainty, including publications in CIA's Studies in Intelligence.

EDUCATION

Ph.D., Quantitative Psychology — The Ohio State University

Master of Information and Data Science — University of California, Berkeley

M.A., Industrial & Organizational Psychology — George Mason University

B.S., Behavioral Sciences & Leadership — United States Air Force Academy

CERTIFICATIONS & ADVANCED TRAINING

Carnegie Mellon University — Agentic AI: Building Autonomous Systems for Real-World Applications (Expected July 2026)

Certified Anti-Fraud Specialist (ACAMS)